

# PARTH BALAR

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## EXPERIENCE

### DevOps · Citta Solutions · Plano, TX

June 2025 – Present

- Architected CI/CD pipelines (GitHub Actions + Terraform) for blue-green/canary deployments on AWS EKS, cutting release cycle time by 35%; integrated ML-based anomaly detection into deployment gates to auto-rollback on predicted SLO breaches.
- Built an LLM-powered on-call assistant (LangChain + Claude API) that triages PagerDuty alerts, queries Splunk/CloudWatch context, and auto-generates incident summaries reducing manual triage time by 50% and MTTR by 45%.
- Developed AI-driven capacity forecasting models (Prophet + Python) on CloudWatch/Prometheus time-series data to predict traffic spikes 24 hrs. ahead, enabling proactive scaling and 18% cloud cost reduction.
- Authored Terraform IaC modules (EKS, RDS, VPC, IAM) as team standards, cutting provisioning from days to under 2 hours; embedded ChromaDB-backed RAG runbook retrieval into postmortem workflows for instant remediation suggestions.

### Site Reliability Engineer (Contract) · Splunk · Ahmedabad, India

Jun 2022 – Dec 2023

- Owned end-to-end SLO/error-budget program for Splunk Cloud across distributed systems at scale, maintaining 99.9% uptime through proactive alerting and automated Python/Shell remediation.
- Built GitOps CI/CD pipelines with Jenkins and Terraform enabling zero-downtime blue-green releases on Kubernetes (AWS & GCP), cutting deployment cycle time by 40%.
- Authored reusable Terraform IaC modules adopted as the team standard for all new service onboarding, cutting environment provisioning time by 60% and scaling solutions fleet-wide.
- Rearchitected indexing/ingestion pipelines, reducing query latency by 30% and ingestion overhead by 20%; built Splunk and Grafana dashboards surfacing degradation before user impact.
- Led blameless postmortems and chaos-engineering reviews across service owners; converted findings into automated recovery runbooks that cut MTTR by 40% over six months.
- Drove FinOps initiative (AWS/GCP migration + rightsizing) via capacity modeling, achieving 15% reduction in cloud spend.

### Site Reliability Engineer Intern · Crest Data · Ahmedabad, India

Dec 2021 – May 2022

- Containerized legacy services with Docker and deployed on Kubernetes (AWS); automated infrastructure provisioning via Terraform + Jenkins, reducing deployment setup from hours to minutes at scale.
- Built an internal automation platform (FastAPI + React) replacing manual data-processing workflows, delivering 60% reduction in manual effort and 40% fewer data errors, eliminating SRE toil.
- Introduced SLI/SLO-based observability using Grafana and Splunk, establishing postmortem culture from scratch; reduced MTTR by 35% within the first quarter.

## PROJECTS

### FRIDAY, Autonomous AI Assistant with Full Computer Control · Python · FastAPI · LangGraph · LangChain · React · ChromaDB · Docker · WebSocket

- Architected a full-stack AI assistant (Python 3.13 + FastAPI backend, React 18 frontend) powered by Claude Sonnet 4.6 with 112+ LangChain tools across 36 modules; designed a LangGraph ReAct autonomous agent with multi-step planning and a two-tier intent router (regex <1ms + LLM fallback) handling 55+ intents across email, file I/O, browser automation, and system control.
- Implemented semantic memory with ChromaDB, a Markov-chain predictive prefetch engine, and a nightly self-reflection loop (Claude Haiku) that evolves the system prompt based on user correction patterns; engineered a 5-source parallel job-search pipeline (ThreadPoolExecutor) with yfinance company-health scoring and SQLite-backed application tracking, deployed via Docker Compose with Nginx reverse proxy.

### Graph-Enhanced RAG Retrieval System · Python · sentence-transformers · ChromaDB · Knowledge Graph · BFS · Cosine Similarity

- Built a hybrid retrieval pipeline combining dense vector search (all-MiniLM-L6-v2 embeddings) with BFS knowledge-graph traversal to surface structurally related code chunks invisible to keyword search.
- Pre-computed L2-normalised entity embeddings enabling single matrix-multiply similarity scoring across 80+ entities; merged graph traversal chunks with standard vector results for up to 50 deduplicated hits per query.

### DataZen, Self-Serve Analytics Platform · Python · React · ML · Data Visualization

- Designed a platform transforming raw CSV uploads into business insights in under 1 minute with auto-generated visualizations, RFM customer segmentation, real-time inventory alerts, and AI-generated executive summaries.

## EDUCATION

### M.S. Information Technology Management · The University of Texas at Dallas

Aug 2023 – May 2025

### B.S. Information Technology · Dharmsinh Desai University, India

Jun 2018 – May 2022

## TECHNICAL SKILLS

**Languages:** Python, Shell Scripting, JavaScript, SQL, Java, C/C++

**Cloud & Infra:** AWS (primary), GCP · Kubernetes, Docker, ECS · Terraform, Ansible

**SRE & Reliability:** SLO/SLI/Error Budget, Chaos Engineering, Incident Management, Blameless Postmortems

**Observability:** Splunk (Cloud & Enterprise), Grafana, Prometheus, ELK Stack, CloudWatch, PagerDuty

**CI/CD & GitOps:** Jenkins, GitHub Actions, GitLab CI/CD · Blue-Green & Canary Deployments

**AI/ML Frameworks:** LangGraph, LangChain, LangChain Anthropic, ChromaDB, sentence-transformers

**Databases:** PostgreSQL, MySQL, MongoDB, SQLite, ChromaDB (vector)

**Web & APIs:** FastAPI, Flask, Django, React.js, Node.js